Data created by kuldeep singh (20/06/2026)

- PAGE 1: WHEAT (ADVANCED DATA)

Crop Name: Wheat (Triticum aestivum)

- **Soil Requirements**

Wheat performs best in **well-drained loamy or clay loam soils** with good fertility. Soil should have adequate organic matter for better root development.

- Ideal pH: 6.0–7.5
- Poor drainage leads to **root rot and reduced yield**
- Deep plowing improves soil aeration

- **Climate**

Wheat is a **rabi (winter) crop** and requires:

- Cool conditions during vegetative growth
- Warm and dry conditions at maturity
- Frost during early stages can damage the crop

- **Irrigation**

Wheat requires **4–6 irrigations** depending on soil:

- Critical stages:
    - Crown root initiation (most important)
    - Tillering ○ Flowering ○ Grain filling
- Over-irrigation can cause **lodging (plants falling down)**

- **Fertilizer**

- Nitrogen improves leaf and stem growth
- Phosphorus helps root development
- Potassium increases disease resistance

Split nitrogen into 2–3 doses for better absorption

- **Diseases & Management**

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- **Rust:** spreads in humid weather → use resistant varieties
- **Smut:** controlled by seed treatment

  Crop rotation reduces disease risk

- **Yield & Harvesting**
  - Average yield: 3–5 tons/hectare
  - Harvest when moisture content is low
  - Delay can cause **grain loss due to shattering**

- PAGE 2: RICE (ADVANCED DATA)

Crop Name: Rice (Oryza sativa)

- **Soil Requirements**

Rice grows best in **clayey and water-retentive soils**.

- Fields should be properly leveled
- Soil puddling helps retain water and control weeds

- **Climate**
  - Requires **high humidity and rainfall**
  - Sensitive to temperature:
    - Low temp → slow growth
    - High temp → poor grain filling

- **Irrigation & Water Management**
  - Requires **standing water (2–5 cm)** during most stages
  - Water management stages:
    - Nursery stage
    - Transplanting stage
    - Tillering stage
  - Drain water 10–15 days before harvest

- **Fertilizer**
  - Nitrogen is most important but excess causes **lodging + pests**

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- Use organic manure + chemical fertilizers

  Zinc deficiency is common → apply zinc sulfate

-Diseases & Pests

- **Blast disease:** fungal, spreads fast
- **Bacterial leaf blight:** water-borne
- **Brown planthopper (pest):** damages plants

  Integrated Pest Management (IPM) is recommended

- **Yield & Harvesting**

- Yield: 4–6 tons/hectare
- Harvest when 80% grains mature
- Proper drying reduces storage loss

- PAGE 3: TOMATO (ADVANCED DATA)

Crop Name: Tomato (Solanum lycopersicum)

- **Soil Requirements**

- Best soil: **Sandy loam with good drainage**
- Soil rich in organic matter improves fruit quality
- Avoid acidic or saline soils

- **Climate**

- Ideal temperature: 20–25°C
- High temperature → flower drop
- Low temperature → slow growth

- **Irrigation**

- Requires **consistent moisture**
- Drip irrigation is best:
  - Saves water
  - Reduces disease
- Water stress causes **fruit cracking**

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- **Fertilizer**

  Balanced NPK required

  - Calcium is important → prevents **blossom end rot**
  - Excess nitrogen → more leaves, fewer fruits

- **Diseases & Pests**

  - **Early blight:** brown spots
  - **Leaf curl virus:** spread by whiteflies
  - **Aphids & whiteflies:** common pests

  Use neem oil or biopesticides for control

- **Yield & Harvesting**

  - Yield: 20–30 tons/hectare
  - Harvest at different stages: ○ Green (transport) ○ Red (local market)
  - Proper handling increases shelf life